Typology of Enthymemes

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1 Introduction

Argumentation entails creating arguments both in favor of and against conflicting assertions. Computational argumentation encompasses various methods for analyzing and reasoning upon arguments and their interconnections. Specifically, **Argument Mining** pertains to the research of automatically identifying and classifying argument structures within text. While this field predominantly focuses on extracting explicit argument structures — comprising **claims** and **premises** linked by **support** and **attack** relations — a more challenging task lies in extracting implicit argument structures within text (e.g. **enthymemes**). Typically, most enthymemes in natural language are understandable to humans. Nevertheless, computational approaches face limitations due to their lack of inherent reasoning abilities and commonsense knowledge. While some strategies to tackle this challenge have been proposed (e.g. using similar but complete arguments to detect and reconstruct enthymemes, incorporating commonsense knowledge relations to facilitate incomplete argument reconstruction etc.), there remains ample room for exploration and development of new methods. These guidelines propose a typology of enthymemes to further facilitate reconstruction of implicitness. In the following, we briefly describe the main concepts of argumentation needed for this annotation (Section 2), then, we present the typology of enthymemes and illustrate it with examples (Section 3). Section 4 describes datasets selected for the annotation of enthymemes typology.

2 Main Argumentation Concepts

An argument is an assertion that necessarily involves a claim. **A claim** is the conclusion of the argument the speaker wishes the audience to believe. A claim may be challenged, therefore, it is important to justify it. In other words, it is necessary to have **an evidence** (**premise**) which supports a claim (Toulmin, 2003). However, one text may contain several claims, some of which are justified by evidences (premises), but others are objected by the latter. Thus, we can establish two kinds of **relation** between a claim and a premise: a premise either **supports** or **attacks** a claim. **An enthymeme** is an incomplete argument which contains hidden assumptions, implicit components and relations that extend beyond the explicit context of an argument (Walton et al., 2008). An **implicit** component in its turn conveys tacit or unspoken meaning, whereas an **explicit** component denotes an overt and directly stated assumption.

3 Types of Enthymemes and their Properties

The necessity to characterize implicitness in argumentation for in-depth analysis and enthymeme reconstruction is highlighted in various research papers (Becker et al., 2019; Hulpus et al., 2019; Saadat-Yazdi et al., 2023). While some authors propose to rely on argumentation models and surrounding context to identify and reconstruct enthymemes, others introduce general knowledge presumed to be missing. Yet another group leverages the power of engineering techniques and large language models for the purpose of reconstruction. However, the importance to describe possible types of implicit components to facilitate their identification and reconstruction is only briefly addressed (Hulpus et al., 2019). Authors suggest that the reconstruction process may vary depending on whether the implicit content involves factual knowledge or subjective, opinionated information. Thus, we decide to delve deeper into the probable typology of enthymemes.

To define potential types of enthymemes, we investigate the pragmatic dimension of implicit statements in argumentation, alongside computational approaches and philosophical perspectives on the matter. Initially, the prevalent form of implicitness arises when individuals deliberately omit information universally acknowledged or easily inferred from the context, aiming to reduce cognitive load in processing such information. This principle corresponds to the Maxim of Quantity of Grice (Grice, 1975): ”Make your contribution as informative as is required. Do not make your contribution more informative than is required.” This pragmatic rule discourages a speaker from making explicit statement of widely accepted facts, therefore, creating one type of implicitness (Razuvayevskaya and Teufel, 2017). Lombardi Vallauri (2016); Lombardi Vallauri et al. (2022) describe a related type of implicitness – implicatures. Implicatures are content which is inferable from the utterance and its context, though not explicitly expressed. The aim of implicatures is to save the audience the effort to process well-known or clear information. In this work, we are going to use one term for this type of implicitness – implicature:

- **Implicature** is content which represents universal knowledge or which is inferable from the utterance and its context, though not explicitly expressed.

To illustrate this type, we will start with the easiest examples comprising short stances that clearly depict an implicature.

1. *Better put some sunscreen on before you go.*

2. *a. Are you going to Bengasi next summer? b. You know, Lybia is very dangerous, presently... (Lombardi Vallauri, 2016)*

Regardless of the fact that the first phrase is not surrounded by any context, it is quite clear that the author implies hot and sunny weather outside, so the risk of getting sunburned is high. However, these circumstances are not explicitly stated, they are rather inferred from the phrase. On the other hand, to understand the second micro-dialog we *i)* need to share some background knowledge (Bengasi is a city in Lybia) and *ii)* we need to infer from the context that the answer *b.* means that they are not going. Both examples either represent implicit information that is shared by communicators or hidden meanings that are inferable from a given text, therefore they both contain **Implicatures**.

To demonstrate implicatures in real discourse, we take examples from Change My View dataset (Tan et al., 2016), a dataset created on the basis of an online community where people present their opinion and reasoning over a topic in question and invite others to challenge their point of view:

3. *[**claim:** Your stance relies on the assumption that religion has no influence on the actions of its followers beyond the superficial.] [**premise:** Yet something must exist that allows this pattern to occur.] I'll narrow it down to religion or culture. So, [**premise:** you are correct if you assume the culture dominates the religion, and you are incorrect if the reverse is true.] [**claim:** With this in mind, I think it's safe to assume the truth is somewhere in between, with both the religion and the culture somehow influencing the unrest we see.]*

Since the title of the discussion is *"Religion is not violent or not violent, its followers are"*, we understand that the author of this post and its commentors will oppose each other on the topic of violent influence of religion. Thus, the last utterance (*"I think it's safe to assume the truth is somewhere in between, with both the religion and the culture somehow influencing the unrest we see*") implicitly states that both religion and culture influence violent behaviour. We categorize such type of implicitness as **Implicature** because humans are able to make this inference due to the topic and context of the passage. For now, we do not take into account other types of implicitness that may be present in this example.

Let's consider another example from the same dataset. The title of the discussion is the same, *"Religion is not violent or not violent, its followers are"*:

4. *So recently I've seen a lot of posts condemning Islam as a violent religion or a sexist religion. [**claim:** I point out that many Christians follow the bible which has numerous examples of sexism, but in application, there are numerous branches of Christianity that are no more sexist than secular groups.] [**premise:** For example, Congregationalists and Universaliists.]*

This example contains implicitness in the ultimate phrase ("*For example, Congregationalists and Universaliists.*"). It implies that both mentioned groups are first religious, second, that they do convey sexism, but no more than secular groups. It is **Implicature** because the context presupposes understanding of all hidden meanings while some background knowledge about branches of Christianity might enhance comprehension of the statement.

Yet another example from the same dataset. The title of this discussion is "Me joining the army is not a waste of time nor does it conflict with my personality":

5. *Other things that I've considered: [**claim:** I love the idea of a brotherhood but I don't agree with the ideology,] [**claim:** the military putting me through Med school will help me jump start life,] [**claim:** I'm very inexperienced and haven't seen much of the world and this I think will help me.]*

The first claim *"I love the idea of a brotherhood but I don't agree with the ideology"* contains **Implicature**: we understand that brotherhood is a characteristic of military service and that military service has its ideology, we understand that this ideology does not correspond to authors preferences and expectations, but without any knowledge about military ideology we cannot imagine what the author exactly means.

Another type of implicitness described by Lombardi Vallauri (2016); Lombardi Vallauri et al. (2022) is vague and ambiguous expressions. Vague and ambiguous expressions require extra information or subjective guessing of the audience as they potentially refer to various entities or states of affairs. Razuvayevskaya and Teufel (2017) describe a related type of implicitness: implicitness of information necessary for understanding, the lack of which arouses cognitive interest of readers and creates ambiguity. Taking into account both approaches, we establish the next type of implicitness:

- **Ambiguity** refers to content that has the potential to denote multiple entities, requiring precision due to its inability to be inferred from the context alone.

This type significantly differs from implicature by its main feature: a single meaning cannot be grasped from the context, moreover, different kinds of audience may interpret the same text differently, perceiving various possible meanings. There might be several possible aims of such type of implicitness. First, ambiguity evokes interest of readers and engages them in communication. Second, ambiguity obscures original meanings, making the audience more compliant as they adopt interpretations that align with their own interests. Third, ambiguity allows the author to hide their defeasible claims and evidences. To illustrate ambiguity, we will first present simple clear examples, where we obviously see that implicitness creates a certain level of ambiguity:

6. *In the library I saw an interesting volume. I didn't take the book because it was valuable.*

7. *She participated at several interviews. They didn't hire her because she's experienced.*

In the example 6 it is not clear if the author didn't take the book because they didn't want to risk damaging it or because a lot of other people might need it. The context given doesn't help solving this ambiguity, so the perception of readers may differ. In the example 7 readers' understanding of the second phrase may also vary: either they didn't hire her at all because they found her overqualified or they hired her but for another reason.

Unfortunately, real data is never so straightforward. Let's consider examples from ElecDeb60to20 dataset (Goffredo et al., 2023), a dataset created on the basis of the US presidential debate transcripts from debates.org:

8. *MR. KENNEDY: Yes. [**claim:** We have been wholly indifferent to Latin America until the last few months.] [**premise** The program that was put forward this summer, after we broke off the sugar quota with Cuba, really was done because we wanted to get through the O.A.S. meeting a condemnation of Russian infiltration of Cuba.] [**premise** And therefore we passed an authorization – not an aid bill – which was the first time, really, since the Inter-American Bank which was founded a year ago was developed, that we really have looked at the needs of Latin America; that we have associated ourselves with those people.]*

Not taking into consideration implicatures that need to be restored for full understanding of the speech, this passage arises ambiguous insights: all the actions taken for Cuba have rather been driven by political expediency than a genuine commitment to improving relations (*"we wanted to get through the O.A.S. meeting a condemnation of Russian infiltration of Cuba"*), but at the same time the author states that the needs of people in Latin America have been taken into consideration (*"we really have looked at the needs of Latin America; that we have associated ourselves with those people"*), so we may perceive *"the program"* as a double-edged sword or we may consider that the author mentions different efforts. Therefore, the type of implicitness present in this utterance is **Ambiguity**.

Let's consider one more example. It is extracted from Change My View dataset (Tan et al., 2016). The title of this argument is *"Surplus Value Theorem is definitive proof that capitalism is an inefficient system"*:

9. *[**claim:** This is one of those views that likes to view production, capital, labor, and all of economics as a zero sum game.] [**claim:** It's not.] [**claim:** Money circulates multiple times in a system, and value is created at each stage.] [**claim:** I.e. you're simplifying it down to a level where it doesn't show all of the uses of the money that actually happen.]*

The context surrounding this passage provides us with all the factual information (for the sake of space we do not include the entire dialog here). However, some parts remain ambiguous: value of money created at different stages may be interpreted differently by experts and non-experts. It seems that this message cannot have the only possible interpretation, therefore, we categorise it as **Ambiguity**.

Both types of implicitness described and illustrated so far belong to an umbrella term **Implicitness of Content**. Its counterpart is **Implicitness of Responsibility** that represents speaker's implicit assumption of responsibility for the content and is represented by presupposition (Lombardi Vallauri, 2016):

- **Presupposition** consists in presenting some notion as already shared by the author and their addressee(s).

Presuppositions can function as effective persuasive devices: while implicatures partially conceal true information to sway the addressee(s) into acceptance, presuppositions openly convey this information, but in a way like it is already known and accepted by the addressee(s). So, rather than creating a scenario where the speaker wants the addressee to believe something, presuppositions establish a scenario where the addressee is assumed to already know and agree upon that something. (Lombardi Vallauri, 2016). Therefore, the probability for doubtful content to be unveiled and processed is reduced. Furthermore, presuppositions can serve to save the addressee(s)' attention and effort over truly familiar content. Lastly, presuppositions can also convey less crucial information without hindering the comprehension of the overall message. It has to be also highlighted that presuppositions are rarer and harder to notice in comparison to implicatures and ambiguities. Let's consider some examples to illustrate this type of implicitness:

10. *Bob has been unmasked. Sue won't forgive his cheating. (Lombardi Vallauri, 2016)*

Here the idea that Bob was doing something wrong is presupposed by the verb *"unmasked"*. Next, the sentence *"Sue won't forgive his cheating."* presupposes that he cheated on her. These meanings are not hidden, but they are presented as if we were all familiar with them.

Let's consider an example from Change My View dataset (Tan et al., 2016) with the title " Freedom of speech is being taken too far" :

11. *[**claim** I certainly value our free speech] but to me [**claim** there is a clear line between exercising your first amendment right ([**premise** "President Obama sucks!"]) and doing things that are known to be offensive to other cultures.] ([**premise** Satirical cartoons of prophets, assassinating leaders, etc.])*

In the example in parallel with an Implicature that introduces the "*first amendment right*" we encounter a **Presupposition** hidden in "*President Obama sucks!*". The idea that no administration has more thoroughly undermined freedom of speech than that of President Obama and that these actions could have been harmful if fully-fledged is presented as presupposed by the premise "*President Obama sucks!*". Even if the readers do not share exactly this knowledge about Obama's administration, it gets quite clear reading the entire passage that Obama tried to change the first amendment right in an unpleasant way. Also, this phrase is made in a way that we do not want to question it as if we share the same background with the author.

Yet another example of a **Presupposition** from the same dataset with the title "Swimming is the best form of exercise" :

12. *a. Swimming is: [**premise** - resistance in many muscle groups, especially when strokes in question are varied and tools such as fins or kickboards are added...]*

    *b. [**premise** Eventually though you'll stop making strength gains and simply be increasing your endurance.] [**claim** This is the fundamental problem with all body-weight exercises.]*

This example represents a small dialog: a commenter refutes a statement about swimming made by an author. The answer of the commenter is a **Presupposition**, as the first phrase (*"Eventually though you'll stop making strength gains and simply be increasing your endurance."*) presupposes that at the beginning swimming may still serve as a means to gain strength. However, swimming is a body-weight exercise, since all body-weight exercises are limited in terms of strength gain progress, swimming is limited too, therefore, strength gain progress will stop at some point. The commenter presents us with their argument as if we shared exactly the same knowledge, so they do not provide us with details either to save our cognitive capacities or probably to avoid our doubts.

4 Dataset

For the annotation of enthymemes types we selected two datasets that *i)* consist of arguments, *ii)* contain implicit argument components. Change My View is the dataset of civil discourse opinions and an homonymous community on Reddit where the data was collected from (Tan et al., 2016). The dataset is built in the following way: the initiator of the discussion creates a title for their post (which contains the major claim of the argument) and then describes the reasons for their belief. Other posters will respond and attempt to change the original poster's view. If they are successful, the original poster will indicate that their view was changed by providing a delta point. Hidey et al. (2017) updated the dataset with manual annotation of argument components (claims and premises) over 39 positive and 39 negative threads, together with annotation of semantic types of premises and claims. Microtext Corpus is our second dataset containing 112 short (3–5 sentence) argumentative texts, professionally translated from German into English, each annotated with argumentation structures, i.e., argument discourse units (ADUs) and their interrelations (Peldszus and Stede, 2013). The texts have straightforward and clear structure, the topics of arguments vary from politics to everyday matters.

4.1 Complementary Instructions

Implicitness and labels for its types concern spans of texts that contain missing (hidden) information or meanings. Hypothetically these spans of texts correspond or partially correspond to claims or premises. In the first annotation round annotators are asked to annotate implicit and explicit spans keeping in mind that implicit entities are self-contained – they are either entire sentences or clauses of sentences:

13. *[I loved The Hobbit, even though it got slow], [but I found Tolkien's prose to be far too slow-paced and filled with inane babbling about things too far removed from the plot.]*

In the example 13 we propose to annotate two spans: full sentence contain several ideas, (presumably) different argument components and does not contain implicitness in all of its parts, therefore, we avoid annotating it entirely with only one label. We cannot consider its separation into smaller spans either, because some of clauses are not self-contained: *"even though it got slow"* does not give us any clear idea about the argument. Thus, it is reasonable to keep two spans and two labels for them.

After selecting implicit and explicit spans, the annotators are asked to label implicit spans with three labels: **Implicature**, **Ambiguity**, **Presupposition**. Therefore, our fine-grained types characterise only implicitness, leaving explicit spans aside. This step is followed by a reconciliation phase where the guidelines may be refined.

A significant challenge in annotating implicitness types lies in the potential overlap between implicit categories. For instance, the same span may simultaneously have characteristics of implicature and ambiguity. To maintain annotation consistency and reduce cognitive load, annotators were instructed to assign the most appropriate label to each span. Allowing multiple labels per instance would increase task complexity for annotators and automatic approaches and potentially decrease inter-annotator agreement. In cases where a span of a different category is embedded within a larger span (e.g., a presupposition inserted in brackets within an ambiguous sentence), both segments were annotated independently with their respective labels.

4.2 Complete Relation Graphs of a CMV and a Microtext arguments

So as to illustrate the complexity of full-text argument analysis, Figure 1 shows a relation graph of a complete CMV dialogue titled *"Religion is not violent or not violent, its followers are"*. For comparison, Figure 2 shows a graph from the Microtext corpus. Both graphs include textual content of argument components and they are linked with support or attack relations.

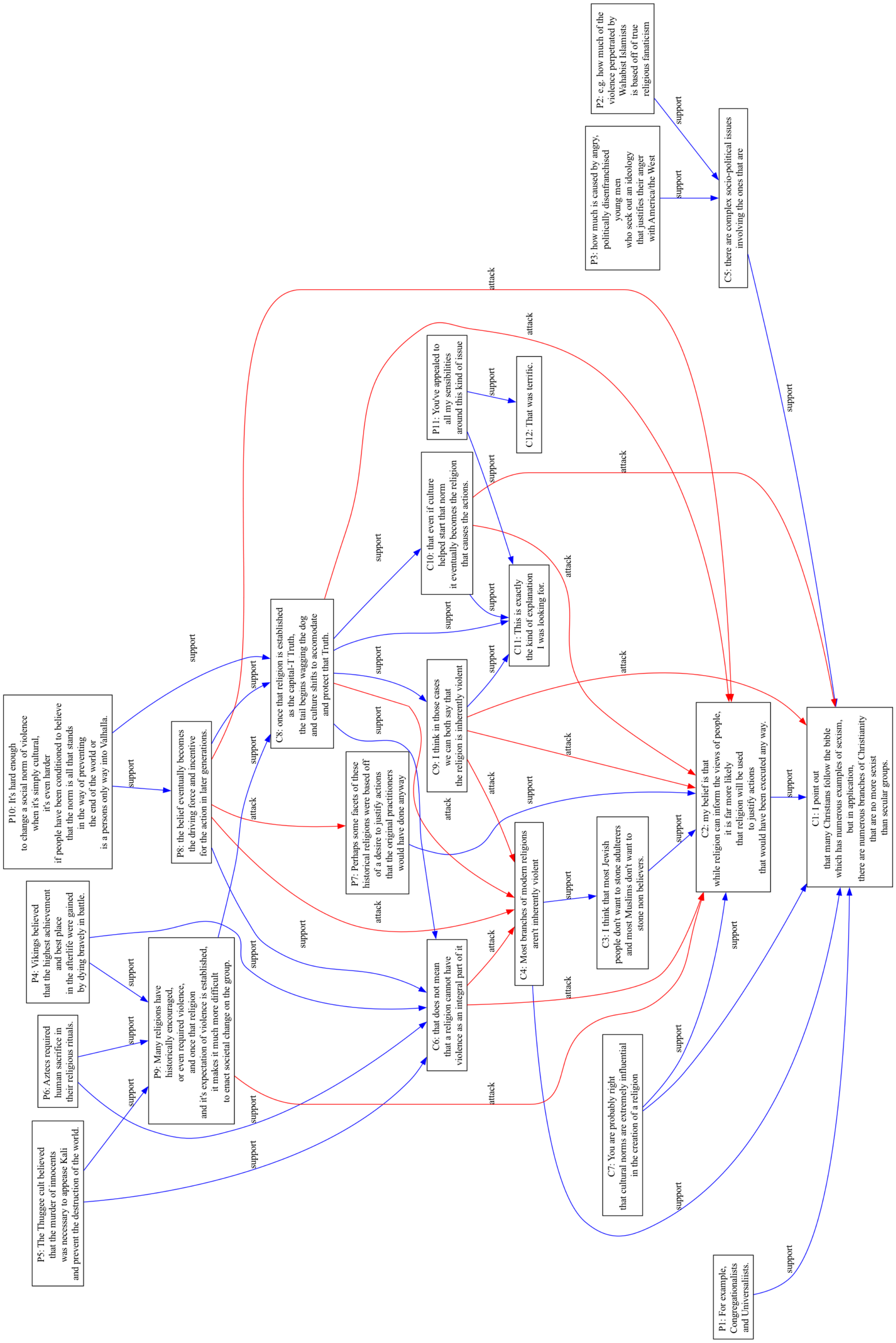

Figure 1: Relation graph of a complete CMV argumentative text.

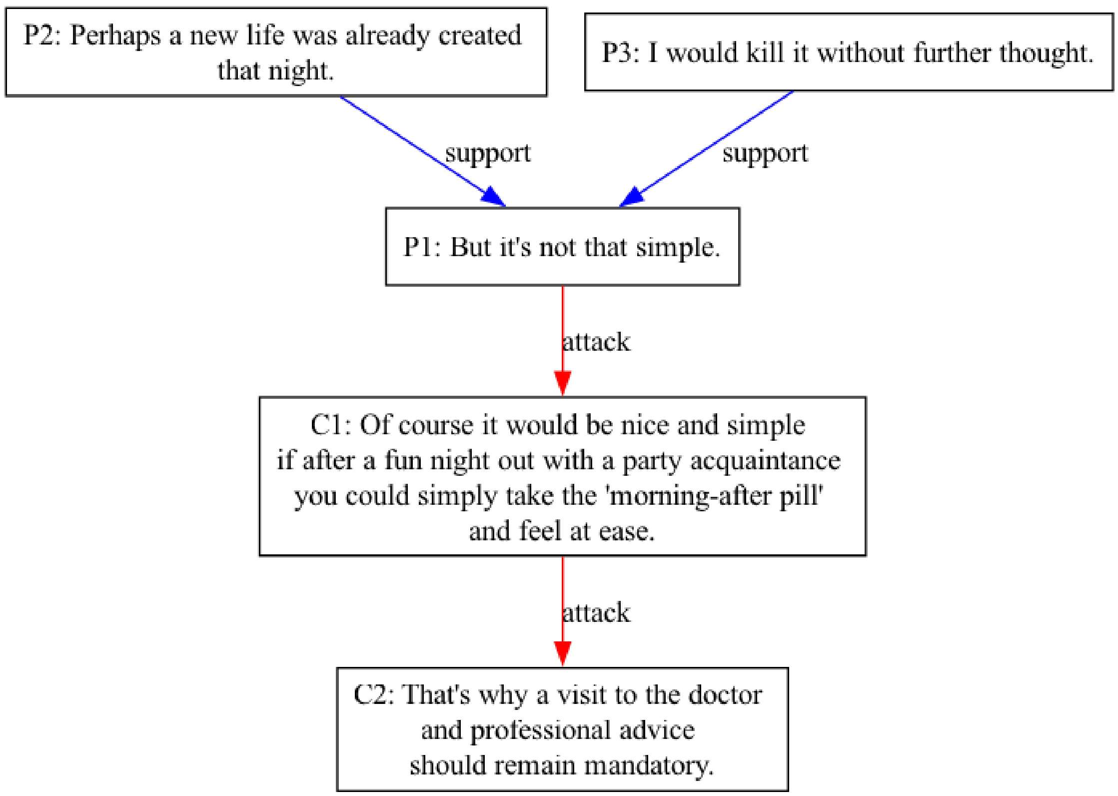

Figure 2: Relation graph of a complete Microtext argument.

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